SELF-RAG: LEARNING TO RETRIEVE, GENERATE, AND CRITIQUE THROUGH SELF-REFLECTION

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ABSTRACT

Despite their remarkable capabilities, large language models (LLMs) often produce responses containing factual inaccuracies due to their sole reliance on the parametric knowledge they encapsulate. Retrieval-Augmented Generation (RAG), an ad hoc approach that augments LMs with retrieval of relevant knowledge, decreases such issues. However, indiscriminately retrieving and incorporating a fixed number of retrieved passages, regardless of whether retrieval is necessary, or passages are relevant, diminishes LM versatility or can lead to unhelpful response generation. We introduce a new framework called **Self-Reflective Retrieval-Augmented Generation (SELF-RAG)** that enhances an LM's quality and factuality through retrieval and self-reflection. Our framework trains a single arbitrary LM that adaptively retrieves passages on-demand, and generates and reflects on retrieved passages and its own generations using special tokens, called *reflection* tokens. Generating reflection tokens makes the LM controllable during the inference phase, enabling it to tailor its behavior to diverse task requirements. Experiments show that SELF-RAG (7B and 13B parameters) significantly outperforms state-of-the-art LLMs and retrieval-augmented models on a diverse set of tasks. Specifically, SELF-RAG outperforms ChatGPT and retrieval-augmented Llama2-chat on Open-domain QA, reasoning and fact verification tasks, and it shows significant gains in improving factuality and citation accuracy for long-form generations relative to these models.[1]

1 INTRODUCTION

State-of-the-art LLMs continue to struggle with factual errors (Mallen et al., 2023; Min et al., 2023) despite their increased model and data scale (Ouyang et al., 2022). Retrieval-Augmented Generation (RAG) methods (Figure 1 left; Lewis et al. 2020; Guu et al. 2020) augment the input of LLMs with relevant retrieved passages, reducing factual errors in knowledge-intensive tasks (Ram et al., 2023; Asai et al., 2023a). However, these methods may hinder the versatility of LLMs or introduce unnecessary or off-topic passages that lead to low-quality generations (Shi et al., 2023) since they retrieve passages indiscriminately regardless of whether the factual grounding is helpful. Moreover, the output is not guaranteed to be consistent with retrieved relevant passages (Gao et al., 2023) since the models are not explicitly trained to leverage and follow facts from provided passages. This work introduces **Self-Reflective Retrieval-augmented Generation (SELF-RAG)** to improve an LLM's generation quality, including its factual accuracy without hurting its versatility, via on-demand retrieval and self-reflection. We train an arbitrary LM in an end-to-end manner to learn to reflect on its own generation process given a task input by generating both task output and intermittent special tokens (i.e., *reflection tokens*). Reflection tokens are categorized into *retrieval* and *critique* tokens to indicate the need for retrieval and its generation quality respectively (Figure 1 right). In particular, given an input prompt and preceding generations, SELF-RAG first determines if augmenting the continued generation with retrieved passages would be helpful. If so, it outputs a **retrieval** token that calls a retriever model on demand (Step 1). Subsequently, SELF-RAG concurrently processes multiple retrieved passages, evaluating their relevance and then **generating** corresponding task outputs (Step 2). It then generates critique tokens to **criticize** its own output and choose best one (Step 3) in terms of factuality and overall quality. This process differs from conventional RAG (Figure 1 left), which

[1]Our code and trained models are available at https://selfrag.github.io/.

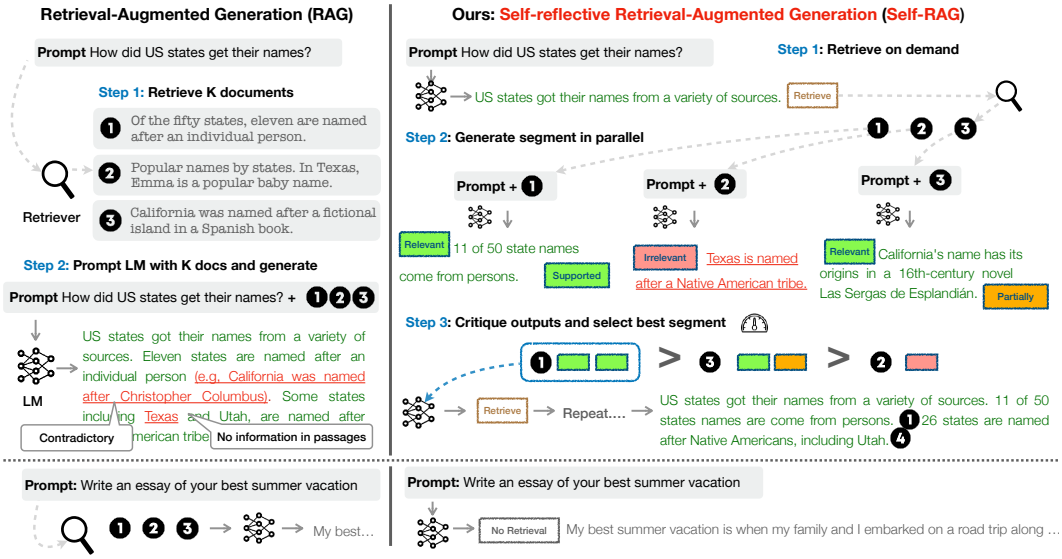

Figure 1: Overview of SELF-RAG. SELF-RAG learns to retrieve, critique, and generate text passages to enhance overall generation quality, factuality, and verifiability.

consistently retrieves a fixed number of documents for generation regardless of the retrieval necessity (e.g., the bottom figure example does not require factual knowledge) and never second visits the generation quality. Moreover, SELF-RAG provides citations for each segment with its self-assessment of whether the output is supported by the passage, leading to easier fact verification.

SELF-RAG trains an arbitrary LM to generate text with reflection tokens by unifying them as the next token prediction from the expanded model vocabulary. We train our generator LM on a diverse collection of text interleaved with reflection tokens and retrieved passages. Reflection tokens, inspired by reward models used in reinforcement learning (Ziegler et al., 2019; Ouyang et al., 2022), are inserted offline into the original corpus by a trained *critic* model. This eliminates the need to host a critic model during training, reducing overhead. The critic model, in part, is supervised on a dataset of input, output, and corresponding reflection tokens collected by prompting a propriety LM (i.e., GPT-4; OpenAI 2023). While we draw inspiration from studies that use control tokens to start and guide text generation (Lu et al., 2022; Keskar et al., 2019), our trained LM uses critique tokens to assess its own predictions after each generated segment as an integral part of the generation output.

SELF-RAG further enables a customizable decoding algorithm to satisfy hard or soft constraints, which are defined by reflection token predictions. In particular, our inference-time algorithm enables us to (1) flexibly adjust retrieval frequency for different downstream applications and (2) customize models' behaviors to user preferences by leveraging reflection tokens through segment-level beam search using the weighted linear sum of the reflection token probabilities as segment score.

Empirical results on six tasks, including reasoning and long-form generation, demonstrate that SELF-RAG significantly outperforms pre-trained and instruction-tuned LLMs that have more parameters and widely adopted RAG approaches with higher citation accuracy. In particular, SELF-RAG outperforms retrieval-augmented ChatGPT on four tasks, Llama2-chat (Touvron et al., 2023) and Alpaca (Dubois et al., 2023) on all tasks. Our analysis demonstrates the effectiveness of training and inference with reflection tokens for overall performance improvements as well as test-time model customizations (e.g., balancing the trade-off between citation previsions and completeness).

2 RELATED WORK

Retrieval-Augmented Generation. Retrieval-Augmented Generation (RAG) augments the input space of LMs with retrieved text passages (Guu et al., 2020; Lewis et al., 2020), leading to large improvements in knowledge-intensive tasks after fine-tuning or used with off-the-shelf LMs (Ram et al., 2023). A more recent work (Luo et al., 2023) instruction-tunes an LM with a fixed number

of retrieved passages prepended to input, or pre-train a retriever and LM jointly, followed by few-shot fine-tuning on task datasets (Izacard et al., 2022b). While prior work often retrieves only once at the beginning, Jiang et al. (2023) propose to adaptively retrieve passages for generation on top of a proprietary LLM or Schick et al. (2023) train an LM to generate API calls for named entities. Yet, the improved task performance of such approaches often comes at the expense of runtime efficiency (Mallen et al., 2023), robustness to irrelevant context (Shi et al., 2023), and lack of attributions (Liu et al., 2023a; Gao et al., 2023). We introduce a method to train an arbitrary LM to learn to use retrieval *on-demand* for diverse instruction-following queries and introduce controlled generation guided by reflections tokens to further improve generation quality and attributions.

Concurrent RAG work. A few concurrent works[2] on RAG propose new training or prompting strategies to improve widely-adopted RAG approaches. Lin et al. (2023) fine-tune both the retriever and LM on instruction-tuning datasets in two steps. While we also train our model on diverse instruction-following datasets, SELF-RAG enables retrieval on demand and selection of the best possible model output via fine-grained self-reflection, making it widely applicable and more robust and controllable. Yoran et al. (2023) use a natural language inference model and Xu et al. (2023) use a summarization model to filter out or compress retrieved passages before using them to prompt the LM to generate the output. SELF-RAG processes passages in parallel and filters out irrelevant ones through self-reflection, without relying on external models at inference. Moreover, our self-reflection mechanism also evaluates other aspects of the model output quality including factuality. LATS (Zhou et al., 2023) prompt off-the-shelf LMs to search for relevant information for question answering tasks and to generate with tree search, guided by LM-generated value scores. While their value function simply indicates an overall score of each generation, SELF-RAG trains to an arbitrary LM to learn to generate fine-grained self-reflection and customizable inference.

Training and generating with critics. Training LLMs with reinforcement learning (e.g., Proximal Policy Optimization or PPO; Schulman et al. 2017) from human feedback (RLHF) has proven effective in aligning LLMs with human preferences (Ouyang et al., 2022). Wu et al. (2023) introduce fine-grained RLHF with multiple reward models. Though our work also studies fine-grained critique on retrieval and generation, we train our target LM on task examples augmented with reflection tokens from a critic model offline, with a far lower training cost compared to RLHF. In addition, reflection tokens in SELF-RAG enable controllable generation at inference, while RLHF focuses on human preference alignment during training. Other works use general control tokens to guide LM generation (Lu et al., 2022; Korbak et al., 2023), while SELF-RAG uses reflection tokens to decide the need for retrieval and to self-evaluate generation quality. Xie et al. (2023) propose a self-evaluation-guided decoding framework, but they focus only on reasoning tasks with one evaluation dimension (reasoning path consistency) and without retrieval. Recent work on LLM refinement (Dhuliawala et al., 2023; Madaan et al., 2023; Paul et al., 2023) prompts a model to generate task output, natural language feedback and refined task output iteratively, but at the cost of inference efficiency.

3 SELF-RAG: LEARNING TO RETRIEVE, GENERATE AND CRITIQUE

We introduce Self-Reflective Retrieval-Augmented Generation (SELF-RAG), shown in Figure 1. SELF-RAG is a framework that enhances the quality and factuality of an LLM through retrieval and self-reflection, without sacrificing LLM's original creativity and versatility. Our end-to-end training lets an LM $\mathcal{M}$ **generate** text informed by **retrieved** passages, if needed, and **criticize** the output by learning to generate special tokens. These *reflection tokens* (Table 1) signal the need for retrieval or confirm the output's relevance, support, or completeness. In contrast, common RAG approaches retrieve passages indiscriminately, without ensuring complete support from cited sources.

3.1 PROBLEM FORMALIZATION AND OVERVIEW

Formally, given input x, we train $\mathcal{M}$ to sequentially generate textual outputs y consisting of multiple segments $y = [y_1, \ldots, y_T]$, where y_t indicates a sequence of tokens for the t-th segment.[3] Generated tokens in y_t include text from the original vocabulary as well as the reflection tokens (Table 1).

[2]All work is arXived within a week of this preprint.

[3]In this paper, we treat one sentence as a segment in our experiments, but our framework is applicable to any segment unit (i.e., sub-sentence).